

p8105_hw2_yg3096

2025-09-30

Problem 1

```
# Set data path
path <- "data/fivethirtyeight_datasets"
pols <- readr::read_csv(file.path(path, "pols-month.csv"),
                        show_col_types = FALSE)

# Clean pols-month
pols_clean <- pols %>%
  # After checking the table, format of mon is "YYYY-M-D"
  separate(mon, into = c("year", "month", "day")) %>%
  mutate(year = as.integer(year), month = as.integer(month),
         month = month.name[month], # Replace with month names
         # After checking the table, there are no null values
         president = case_when(prez_gop == 1 ~ "gop",
                               prez_dem == 1 ~ "dem")) %>%
  select(-prez_gop, -prez_dem, -day)
head(pols_clean)
```

```
## # A tibble: 6 x 9
##   year month   gov_gop sen_gop rep_gop gov_dem sen_dem rep_dem president
##   <int> <chr>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <chr>
## 1  1947 January     23     51    253     23     45    198 dem
## 2  1947 February  23     51    253     23     45    198 dem
## 3  1947 March     23     51    253     23     45    198 dem
## 4  1947 April     23     51    253     23     45    198 dem
## 5  1947 May       23     51    253     23     45    198 dem
## 6  1947 June      23     51    253     23     45    198 dem
```

```
snp <- readr::read_csv(file.path(path, "snp.csv"), show_col_types = FALSE)
# Clean snp
snp_clean <- snp %>%
  # M-D-YY format
  separate(date, into = c("month", "day", "year")) %>%
  mutate(year = as.integer(year),
         # Change 2-digit year format into 4-digit
         year = if_else(year < 50, year + 2000L, year + 1900L),
         month = as.integer(month), month = month.name[month]) %>%
  select(year, month, close)
head(snp_clean)
```

```
## # A tibble: 6 x 3
##   year month   close
```

```
##      <int> <chr>      <dbl>
## 1  2015 July        2080.
## 2  2015 June        2063.
## 3  2015 May         2107.
## 4  2015 April       2086.
## 5  2015 March       2068.
## 6  2015 February   2104.
```

```
unemp <- readr::read_csv(file.path(path, "unemployment.csv"),
                        show_col_types = FALSE)
# Clean unemployment
unemp_long <- unemp %>%
  pivot_longer(cols = Jan:Dec, names_to = "month",
               values_to = "unemployment_rate") %>%
  mutate(year = as.integer(Year),
         # Rename Jan/Feb/... to January/February/... for consistency
         month = month.name[match(month, month.abb)]) %>%
  select(year, month, unemployment_rate)
head(unemp_long)
```

```
## # A tibble: 6 x 3
##   year month      unemployment_rate
##   <int> <chr>          <dbl>
## 1  1948 January          3.4
## 2  1948 February        3.8
## 3  1948 March           4
## 4  1948 April          3.9
## 5  1948 May            3.5
## 6  1948 June           3.6
```

```
# Merge snp into pols
pols_snp <- pols_clean %>%
  left_join(snp_clean, by = c("year", "month"))

# Then merge into unemployment
merged_all <- pols_snp %>%
  left_join(unemp_long, by = c("year", "month"))

# Check result
dim(merged_all)
```

```
## [1] 822  11
```

```
range(merged_all$year, na.rm = TRUE)
```

```
## [1] 1947 2015
```

```
sum(is.na(merged_all$close)) # Check NA values in close column
```

```
## [1] 36
```

```
sum(is.na(merged_all$unemployment_rate)) # Check NA values in unemployment
```

```
## [1] 12
```

The three raw data contains the number of national politicians who are democratic or republican, Standard & Poor's stock market index (S&P) and rate of unemployment by month. After merging these data set by year and month, dimension of result is 822 x 11, year range is `range(merged_all$year, na.rm = TRUE)`. Key variables include year, month, gov_gop, sen_gop, rep_gop, gov_dem, sen_dem, rep_dem.

Problem 2

```
# Set data path
path <- "data/tw.xlsx"
# Read and clean Mr. Trash Wheel sheet, updated
mr <- read_excel(path, sheet = "Mr. Trash Wheel", skip = 1) %>%
  janitor::clean_names() %>% # Clean names
  select(-starts_with("x")) %>% # Delete NA columns
  filter(!is.na(dumpster)) %>% # Omit non-data rows
  # Round number of sports balls to nearest integer
  mutate(year = as.integer(year),
         sports_balls = as.integer(round(sports_balls, 0)),
         wheel = "Mr. Trash Wheel")
```

```
## New names:
## * ' ' -> '...15'
## * ' ' -> '...16'
```

```
# Read and clean Professor Trash Wheel sheet
prof <- read_excel(path, sheet = "Professor Trash Wheel", skip = 1) %>%
  janitor::clean_names() %>% # Clean names
  select(-starts_with("x")) %>% # Delete NA columns
  filter(!is.na(dumpster)) %>%
  # No sports balls column in this sheet
  mutate(year = as.integer(year), wheel = "Professor Trash Wheel")

# Read and clean Gwynnda Trash Wheel sheet
gwynnda <- read_excel(path, sheet = "Gwynnda Trash Wheel", skip = 1) %>%
  janitor::clean_names() %>% # Clean names
  select(-starts_with("x")) %>% # Delete NA columns
  filter(!is.na(dumpster)) %>%
  # Also no sports balls column in this sheet
  mutate(year = as.integer(year), wheel = "Gwynnda Trash Wheel")

trash_all <- bind_rows(mr, prof, gwynnda)
```

After cleaning and combining the three Trash Wheel datasets, the result dataset contains 1033 observations. Key variables include `dumpster` (the dumpster identifier), `weight_tons` (weight of trash collected), `volume_cubic_yards` (volume of trash) and counts of specific items such as

plastic_bottles, polystyrene and cigarette_butts. For Mr. Trash Wheel, the dataset also includes sports_balls, rounded to the nearest integer. Across all years of available data, Professor Trash Wheel collected a total of 246.74 tons of trash. In June of 2022, Gwynnda collected 1.812×10^4 cigarette butts.

Problem 3

```
# Set data path and read data
path <- "data/zillow_data"
zori <- readr::read_csv(file.path(path,
                                "Zip_zori_uc_sfrcondomfr_sm_month_NYC.csv"),
                      show_col_types = FALSE)

zori_long <- zori %>%
  rename(zip = RegionName) %>%
  mutate(zip = stringr::str_pad(as.character(zip), 5, pad = "0")) %>%
  pivot_longer(cols = matches("^\\d{4}-\\d{2}-\\d{2}$"), # YYYY-MM-DD
               names_to = "date", values_to = "zori") %>%
  mutate(date = ymd(date), year = year(date),
         month = month(date, label = TRUE, abbr = FALSE,
                        locale = "C") %>%
  as.character()) %>%
  select(zip, year, month, date, zori)
dim(zori_long)
```

```
## [1] 17284      5
```

```
head(zori_long)
```

```
## # A tibble: 6 x 5
##   zip   year month   date      zori
##   <chr> <dbl> <chr>   <date>   <dbl>
## 1 11368 2015 January 2015-01-31    NA
## 2 11368 2015 February 2015-02-28    NA
## 3 11368 2015 March   2015-03-31    NA
## 4 11368 2015 April   2015-04-30    NA
## 5 11368 2015 May     2015-05-31    NA
## 6 11368 2015 June    2015-06-30    NA
```

```
zip <- readr::read_csv(file.path(path, "Zip Codes.csv"), show_col_types = FALSE)
zip_lookup <- zip %>%
  janitor::clean_names() %>%
  transmute(zip = str_pad(as.character(zip_code), width = 5,
                        pad = "0"),
            county = county, neighborhood = neighborhood) %>%
  # Divide by borough name
  mutate(borough = case_when(
    str_detect(county, regex("^bronx", ignore_case = TRUE)) ~ "Bronx",
    str_detect(county, regex("^kings",
                            ignore_case = TRUE)) ~ "Brooklyn",
    str_detect(county, regex("^new york",
```

```

                                ignore_case = TRUE)) ~ "Manhattan",
  str_detect(county, regex("^queens",
                                ignore_case = TRUE)) ~ "Queens",
  str_detect(county, regex("^richmond",
                                ignore_case = TRUE)) ~ "Staten Island",
  TRUE ~ NA_character_)) %>%
  select(zip, borough, neighborhood) %>%
  distinct()

# Check result
zip_lookup %>% summarize(n = n(), na_zip = sum(is.na(zip)),
                        na_borough = sum(is.na(borough)),
                        na_neigh = sum(is.na(neighborhood)))

```

```

## # A tibble: 1 x 4
##       n na_zip na_borough na_neigh
##   <int> <int>    <int>    <int>
## 1   322     0        0       142

```

```

# Keep distinct matches only
zip_lookup <- zip_lookup %>%
  distinct(zip, .keep_all = TRUE)
head(zip_lookup)

```

```

## # A tibble: 6 x 3
##   zip   borough neighborhood
##   <chr> <chr>    <chr>
## 1 10451 Bronx   High Bridge and Morrisania
## 2 10452 Bronx   High Bridge and Morrisania
## 3 10453 Bronx   Central Bronx
## 4 10454 Bronx   Hunts Point and Mott Haven
## 5 10455 Bronx   Hunts Point and Mott Haven
## 6 10456 Bronx   High Bridge and Morrisania

```

```

# Merge data
zori_full <- zori_long %>%
  left_join(zip_lookup, by = "zip") %>%
  relocate(zip, borough, neighborhood, year, month, date, zori)

# Basic information of result
n_obs <- nrow(zori_full)
n_zip <- n_distinct(zori_full$zip)
n_neigh <- n_distinct(zori_full$neighborhood, na.rm = TRUE)
list(n_obs = n_obs, n_zip = n_zip, n_neigh = n_neigh)

```

```

## $n_obs
## [1] 17284
##
## $n_zip
## [1] 149
##
## $n_neigh
## [1] 42

```

```

# Find zip in lookup but not in zori
zip_only_in_lookup <- anti_join(zip_lookup %>%
                                select(zip) %>%
                                distinct(), zori_long %>%
                                select(zip) %>%
                                distinct(), by = "zip")

# Check result
nrow(zip_only_in_lookup)

## [1] 171

head(zip_only_in_lookup)

## # A tibble: 6 x 1
##   zip
##   <chr>
## 1 10464
## 2 10474
## 3 10475
## 4 10499
## 5 10550
## 6 10704

# get 2020-01 and 2021-01
zori_jan <- zori_full %>%
  filter(month == "January", year %in% c(2020, 2021)) %>%
  select(zip, borough, neighborhood, year, zori) %>%
  pivot_wider(names_from = year, values_from = zori,
              names_prefix = "y") %>%
  mutate(drop_2020_2021 = y2021 - y2020)

# Find 10 zip with most drop
top10 <- zori_jan %>%
  filter(!is.na(y2020), !is.na(y2021)) %>%
  arrange(drop_2020_2021) %>%
  slice_head(n = 10) %>%
  select(zip, borough, neighborhood, y2020, y2021, drop_2020_2021)

# Show result
knitr::kable(top10, caption = "Largest ZORI drops from Jan 2020 to Jan 2021")

```

Table 1: Largest ZORI drops from Jan 2020 to Jan 2021

zip	borough	neighborhood	y2020	y2021	drop_2020_2021
10007	Manhattan	Lower Manhattan	6334.211	5421.614	-912.5966
10069	Manhattan	NA	4623.042	3874.918	-748.1245
10009	Manhattan	Lower East Side	3406.442	2692.187	-714.2550
10016	Manhattan	Gramercy Park and Murray Hill	3731.135	3019.431	-711.7045
10001	Manhattan	Chelsea and Clinton	4108.098	3397.648	-710.4499
10002	Manhattan	Lower East Side	3645.416	2935.113	-710.3028
10004	Manhattan	Lower Manhattan	3149.658	2443.697	-705.9608

zip	borough	neighborhood	y2020	y2021	drop_2020_2021
10038	Manhattan	Lower Manhattan	3573.201	2875.616	-697.5853
10012	Manhattan	Greenwich Village and Soho	3628.566	2942.344	-686.2218
10010	Manhattan	Gramercy Park and Murray Hill	3697.284	3012.353	-684.9304

The final tidy dataset contains 17284 observations across 149 ZIP codes and 42 neighborhoods. While the borough variable is complete, approximately 142 ZIP codes do not have neighborhood information in the lookup file.

A total of 171 ZIP codes appear in the lookup dataset but not in the Zillow ZORI data. They are likely to be PO Box-only ZIPs, industrial or commercial areas or lacking sufficient rental market coverage. Therefore these ZIPs are excluded from Zillow's rental index.

When comparing rental prices between January 2020 and January 2021, the 10 ZIP codes with the largest declines are shown in the table above. These declines occurred mainly in higher-rent neighborhoods. Notably, one ZIP code lacks neighborhood label in the lookup data, which explains the occasional NA values in the neighborhood column.